4. Estimation of sodium hydroxide

Aim :

To estimate the amount of sodium hydroxide dissolved in 250 ml of the given unknown solution volumetrically. For this you are given with a standard solution of sodium carbonate solution of normality 0.0948 N and hydrochloric acid solution as link solution.

Principle:

Neutralization of Sodium carbonate by HCl is given below. To indicate the end point, methyl orange is used as an indicator.

$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$$

Neutralization of Sodium hydroxide by HCl is given below. To indicate the end point, phenolphthalein is used as an indicator.

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

Short procedure:

s.no	Content	Titration-I	Titration-II
1	Burette solution	HCl (link solution)	HCl (link solution)
2	Pipette solution	20 ml of standard Na_2CO_3 solution	20 ml of unknown NaOH solution
4	Temperature	Lab temperature	Lab temperature
5	Indicator	Methyl orange	Phenolphthalein
6	End point	Colour change from straw yellow to pale pink	Disappearance of pink colour
7	Equivalent weight of NaOH = 40		

Procedure :

Titration–I

(Link HCl)Vs (standard Na_2CO_3)

Burette is washed with water, rinsed with HCl solution and filled with same HCl solution up to the zero mark. Exactly 20 ml of standard Na_2CO_3solution is pipetted out into the clean, washed conical flask. To This solution 2 to 3 drops of methyl orange indicator is added and titrated against HCl link solution from the burette. HCl is added drop wise till the colour change from straw yellow to pale pink. Burette reading is noted and the same procedure is repeated to get concordant values.

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